INFO 4707 · Spring 2026 · University of North Texas

Visualizing International Science and Engineering Fair Trends

2014 – 2024

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14,873
Projects Analyzed

2014–2024
Years Covered

90
Countries

25
ISEF Categories

What is ISEF?

The pinnacle of pre-college science competition worldwide

- The International Science and Engineering Fair (ISEF) is the world's largest international pre-college science competition
- First held in 1950, now sponsored by Regeneron since 2020
- ~1,700 finalists from 75+ countries compete annually after advancing through regional and national selections
- Students submit original research projects with abstracts, compete for $9M+ in prizes and scholarships
- Finalists represent the top ~0.03% of the estimated 7 million students who participate in affiliated local fairs

$9M+
Awarded Annually

75+
Countries

1,700
Finalists / Year

22
Categories

Research Gap

Existing literature focuses on student experience and motivation — not on what topics students choose, how those topics shift over time, or whether topic choice predicts success.

RESEARCH QUESTION

Three Hypotheses Tested

To what extent do patterns of topic popularity in ISEF abstracts correlate with competitive outcomes — and does winning often mean winning big?

H1 AI / ML Rise

AI, machine learning, and computing topics have grown significantly as a share of ISEF submissions from 2014–2024.

H2 Award Independence

Certain ISEF categories receive systematically larger award dollar amounts, regardless of how often they win.

H3 Trend ≠ Win Rate

Projects in categories that have grown in popularity over time win awards at a higher rate.

Data Collection & Cleaning

Kaggle dataset scraped from the Society for Science Abstract Search

Dataset

- Source: Kaggle — "All ISEF Projects" (Kaiser, 2022)
- 14,911 raw project entries from 2014–2024
- Attributes: title, category, year, abstract, country, state, awards
- 38 duplicate projects removed → 14,873 final entries
- Empty title/abstract cells filled with " " (not removed)
- 2015–2016 excluded from category analysis: 68–82% null category rate
- 2020 excluded from award analysis: competition cancelled (COVID-19)
- 2024 excluded from award analysis: data collection gap in source

1 **Award Parsing**

parse_awards() converts raw text to lists; extract_award_money() captures dollar amounts

2 **Tier Assignment**

9-tier system based on percentile-informed breakpoints ($0 → $100k+)

3 **Category Audit**

Null audit, name inventory, presence matrix — identified restructure in 2017

4 **Keyword Search**

5 AI/ML keywords tracked across all abstract text 2014–2024

Award Distribution: Skewed and Zero-Heavy

Most ISEF finalists receive no monetary compensation

71%
Projects win $0

$1,000
Median (winners)

$252K
Maximum award

2-Process
Grand + Special Awards

Key EDA Findings

- Tier 1–3 ($1–$2,000) hold the most winners: 741 / 664 / 504 projects respectively — reflecting ISEF's Fourth, Third, and Second Award structure
- Tier 7 ($25K–$50K) contains only 2 projects — a natural dead zone. No grand award lands here; only rare multi-award stacking reaches this range
- Tiers 8 & 9 ($50K+) are larger than Tier 7, driven by scholarship stacking from special award organizations, not ISEF judging
- 741 winners received non-cash recognition only ($0 monetary value) — award occurrence and award magnitude are separate phenomena
- The two-process model: ISEF grand awards equalize win rates across categories; special awards are sponsor-driven and unevenly distributed

Computing Keywords in ISEF Abstracts

Percentage of all project abstracts containing each AI/ML keyword, by year

Keyword	2014	2017	2019	2021	2024	Change
Machine Learning	0.93%	3.21%	4.84%	7.44%	7.66%	8×
Neural Network	0.79%	2.21%	5.26%	7.37%	7.14%	9×
Deep Learning	0.07%	0.71%	2.03%	3.99%	4.68%	67×
Artificial Intelligence	0.72%	0.71%	1.54%	2.13%	2.68%	4×
Computer Vision	0.29%	0.50%	0.98%	1.38%	1.19%	4×

Every keyword grew. All five independently tracked terms moved in the same direction across 11 years.

3 Growth Phases

2014–2017

Dormant baseline. All keywords below 3.5%

2018–2021

Rapid acceleration as AI tools became accessible

2022–2024

Plateau — AI is now embedded, not novel

RESULT — H1

H1: AI/ML Topics Rose Dramatically ✓

Confirmed — consistent growth across all five keywords

67×
Deep Learning
2014–2024

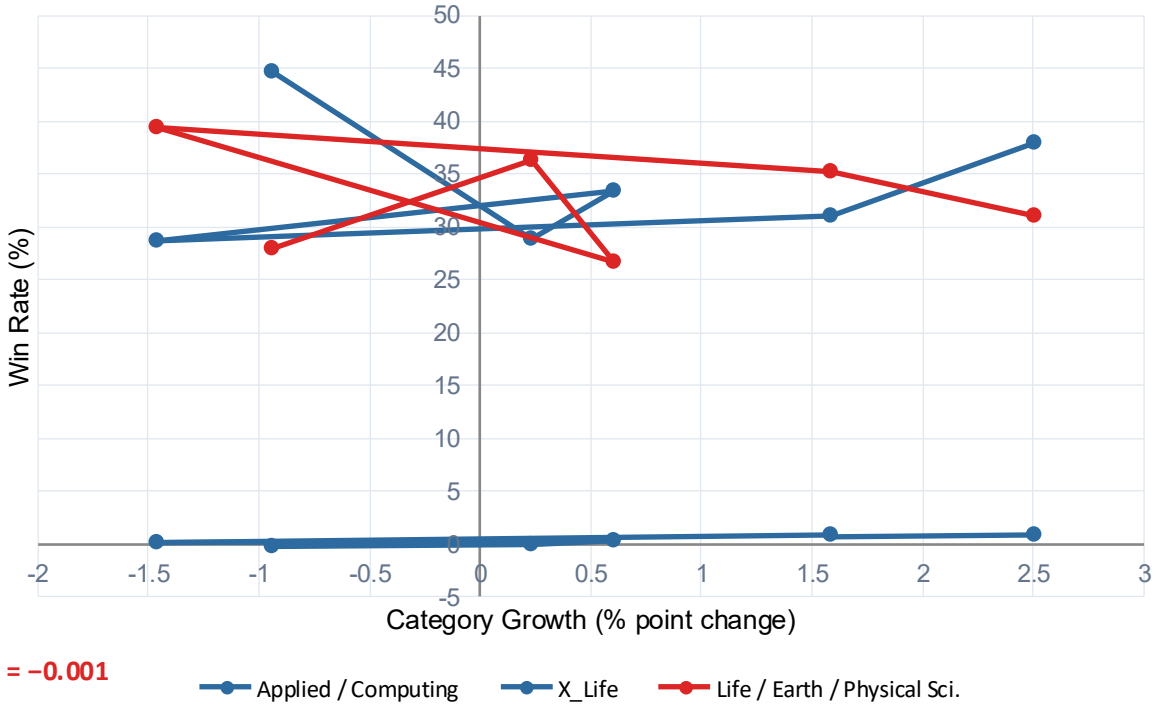
9×
Neural Network
2014–2024

8×
Machine Learning
2014–2024

RESULT — H3

H3: Growth Does Not Predict Win Rate ✗

Refuted — Pearson r = −0.001 ≈ no relationship



r = −0.001

Mathematics

Fastest shrinking: −5.0 pp

Highest win rate: 44.6%

Judges reward rigor, not trend.

Translational Med. Science

Fastest growing: +3.4 pp

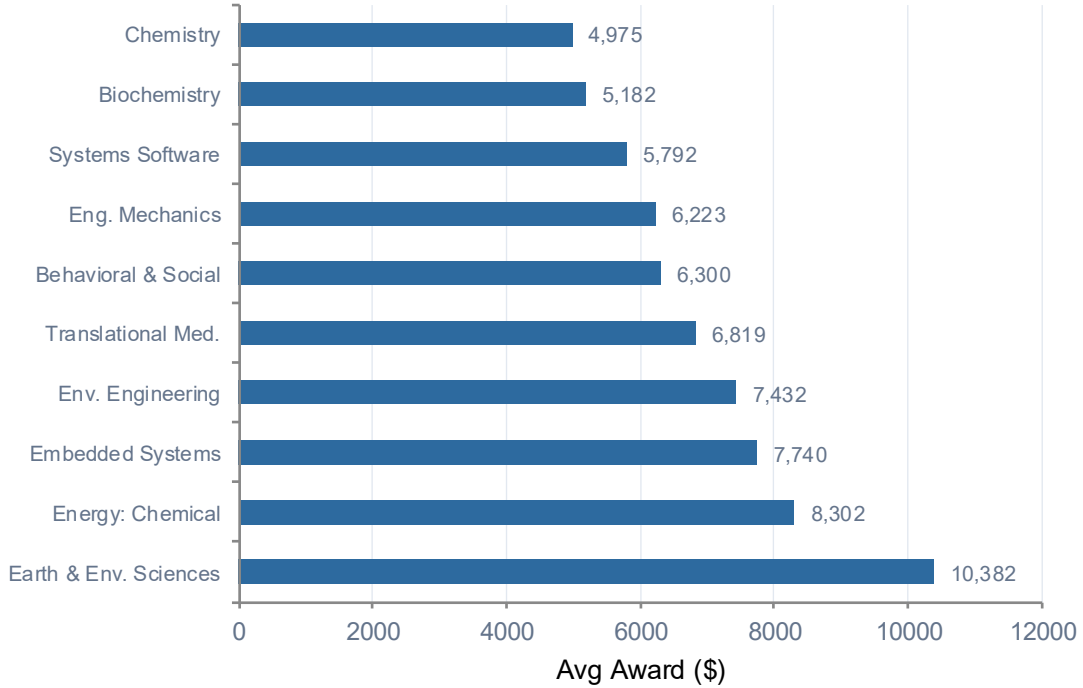
Below avg win rate: 22.9%

Judges reward rigor, not trend.

RESULT — H2

H2: Award $ Independent of Win Rate ✓

Confirmed — Pearson r = 0.142 ≈ no meaningful relationship



The Key Contrast

Mathematics

Win Rate: 44.6% (HIGHEST)

Avg Award: $1,859 (LOWEST)

Fewer special award sponsors in pure math

Earth & Environmental Sci.

Win Rate: 32.7% (below median)

Avg Award: $10,382 (HIGHEST)

Dense sponsor network: gov agencies, conservation funds, corporations

DISCUSSION

Why Are Win Rate and Award $ Independent?

The two-process model explains the structural divergence

Process 1 — ISEF Grand Awards

- Awarded by ISEF judges based on merit
- Distributed proportionally to projects per category — built-in equalization
- Best of Category: $5,000 | 1st: $3,000 | 2nd: $1,500 | 3rd: $1,000 | 4th: $500
- This mechanism ensures all active categories have similar win rates
- Explains why win rates cluster between 22–44% across all 24 categories

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Process 2 — Special Awards

- Awarded by independent sponsor organizations (government, corporate, foundations)
- Sponsors choose which categories to support and how much to offer
- Can include tuition scholarships worth $40K–$180K+
- Environmental, biomedical, and applied categories attract more sponsors
- Pure math and abstract physics attract fewer external sponsors
- Creates the award $ gap between categories that is entirely uncorrelated with win rates

DISCUSSION

What the Findings Mean

H1

AI/ML is now embedded in ISEF

Student research shifted dramatically toward computing and AI over the past decade, reflecting both cultural availability and genuine student interest. By 2024, machine learning appears in 1 in 13 project abstracts.

H3

ISEF judging rewards rigor, not trend

The categories with the highest win rates — Mathematics, Systems Software — share methodological rigor and clear evaluation criteria. ISEF's award system appears insulated from cultural fashion, which has important equity implications.

H2

Financial opportunity is sponsor-driven

A Mathematics winner receives ~$1,859 on average; an Earth & Env. Sciences winner receives ~$10,382. If students are unaware of this gap, science fair may channel financial opportunity by sponsor interest — not competitive merit.

Limitations & Data Gaps

Limitation	Description	Impact
2015–2016 Data Gap	68–82% null category rates from source; both years excluded from category analysis. Abstract text retained for keyword analysis.	HIGH
ISEF 2020 Cancelled	COVID-19 cancellation — no judging or awards. Only 2 apparent "winners" are data artifacts. Excluded from all award analysis.	HIGH
2024 Award Gap	Competition held; awards distributed ($75K top prize to Grace Sun). But award data missing from source. Included in topic analysis only.	MEDIUM
Award Conflation	award_money sums grand awards + special awards. A $1,000 third-place winner who gets a $40K scholarship appears as a high earner.	MEDIUM
Category Restructure	ISEF renamed categories in 2017 and made smaller changes in 2020 and 2022. Broad category mapping attempts to account for this.	MEDIUM
Selection Bias	All 14,873 projects already passed regional/national selection. Findings cannot generalize to lower-level science fairs or general students.	LOW

FUTURE WORK

Extensions & Future Research

Google Trends Integration

Overlay public search interest for AI/ML keywords against ISEF abstract frequency to test whether students lead or lag public interest in scientific topics.

Country & University Rankings

Integrate CWUR university ranking data to test whether national academic infrastructure predicts ISEF win rates — an equity and access question.

ISEF Placement Tier Analysis

Parse award text to extract official placement tiers (1st, 2nd, 3rd) separately from special awards, enabling cleaner tests of judging behavior vs. sponsorship.

Geographic & Equity Analysis

Test whether country or US state of origin predicts topic choice and award outcomes — connecting the disciplinary equity gap to a geographic dimension.

Latent Topic Modeling

Apply LDA topic modeling to uncover hidden themes within categories and track topic-level win rates — finer-grained than the category-level analysis here.

Longitudinal Career Outcomes

Link ISEF finalists to academic and professional records to test whether topic choice or award level predicts the field a student ultimately works in.

Conclusions

H1

CONFIRMED

AI/ML topics grew 8–67× across 2014–2024. The shift is real, accelerating, and consistent.

H2

CONFIRMED

Win rate and award $ are independent (r = 0.14). Sponsor networks — not judges — drive prize magnitudes.

H3

REFUTED

Category growth does not predict win rate (r = −0.001). Judges reward rigor, not trend.

Broader Implication

ISEF's judging rewards scientific depth regardless of whether a topic is trending — but its financial reward structure reflects sponsor priorities that are unevenly distributed across disciplines. Students pursuing mathematics and pure science may win as often as anyone but take home far less. Understanding this structure matters for equitable access to the financial opportunities science fair promises.

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Data: Kaiser, W. (2022). All ISEF Projects. Kaggle.